

- 1) Develop an NLP for design optimization of an engineering component or system. (5)/10
- 2) Classify optimization techniques. (5) Explain the role of optimization in design. (5)
- 3) Discuss the Lagrange's condition for constrained optimum. (5)
- 4) Explain the role of penalty multipliers in interior and exterior penalty function method. (5)/8
- 5) Develop an NLP for multi-objective optimization problem. (5)
- 6) Compare traditional and non-traditional optimization algorithm. (5)
- 7) Discuss the weighted matrices method for multi-objective optimization. (5)
- 8) Explain the ϵ -constrained method for multi-objective optimization. (5)
- 9) Discuss the basic principles of region elimination methods used in optimization techniques. (5) / Distinguish bracketing method and region elimination method. (5)
- 10) Explain the algorithm for Golden section method / Secant method for optimization. (5) / How is secant method used to find an optimum solution?
- 11) Apply Bisection method (two iterations) to find out optimum solution for the following objective function. (5) (10) $\rightarrow f(x) = x^2 - 6x + 10$ (or Golden section method.)
 $f(x) = x^2 + 5x + 3$.
- 12) Why algorithm for single variable objective function is not applicable for multivariable objective function? (5)
- 13) Explain Cauchy's steepest descent method for multivariable optimization. (5)
- 14) Why Marquand method is more efficient? (5)
- 15) Find out the value of x at which the following function is minimum, maximum or point of inflection. (5)
 $f(x) = x^4 - 2x^3 + 6x^2 - 5x + 4$
- 16) Apply interval halving method or Golden section method to find out minimum value of the objective function (2 iterations)
 $f(x) = x^2 + 25/x$. (10)
- 17) Apply Cauchy's steepest descent method for multivariable optimization for the following function, $f(x) = (x_1 - 2x_2 - x_1x_2)^2$ (two iterations) (10)

- 18) Describe optimality criteria for equality constraints (6)
- 19) Explain particle swarm optimization technique (6)
- 20) Explain genetic algorithm used for optimization. (10)
- 21) Explain the algorithm for simulated annealing. (6) Working principle of PSO and simulated annealing (10)
- 22) Explain Ant colony optimization algorithm. (10)
- 23) Describe the working principle of Artificial Neural Network. (10)
- 24) Explain Pareto optimal solution. (5)
- 25) Explain any technique for finding "non dominated" solution. (5)
- 26) Compare feasible design and optimum design. (5)
- 27) How optimization algorithm for multivariable is different from that for single variable objective function. (5)
- 28) Explain the sufficient condition for an optimum point for a funcⁿ. (5)
- 29) Describe the features of the evolutionary algorithms. (5)
- 30) Explain the significance of optimization in design. (5)
- 31) Explain Simplex method for optimization. (4)
- 32) How penalty function method works for constrained optimization. (6)
- 33) What is the role of penalty multipliers. (3)

Classification of Optimization Techniques:-

- ① Classification Based on no. of design variables
 - Single Variable
 - Multi-Variable
- ② Classification Based on the Nature of the design variables
 - Parameter or static optimization problem
 - Trajectory or Dynamic optimization problem
- ③ Based on Physical structure of the problem
 - optimal control problem
 - Non-optimal control problem
- ④ Based on Nature of the equations involved
 - Linear
 - Nonlinear
 - Geometric
 - Quadratic programming
- ⑤ Based on the permissible values of the design variables
 - Integer programming problem
 - Real-Valued programming problem
- ⑥ Based on the Deterministic nature of the variables
 - Deterministic prog. problem
 - Stochastic prog. problem
- ⑦ Based on the separability of the functions
 - separable
 - Non-separable
- ⑧ Based on the no. of objective functions
 - Single-objective programming
 - Multi-objective programming
- ⑨ Based on the constraints
 - Constrained Optimization problem
 - Unconstrained Optimization problem
- ⑩ Based on optimization method
 - Direct Search
 - Gradient Search
- ⑪ Based on nature of solution
 - Local
 - Global
- ⑫ Based on search techniques
 - Traditional
 - Non-traditional

- ① Classical methods of differential calculus for unconstrained maxima and minima of a function of several variables.
- ② For problems with equality constraints $\rightarrow$ the Lagrange multiplier method
- ③ " " " inequality " $\rightarrow$ the Kuhn-Tucker conditions.
- ④ Techniques of linear, nonlinear, geometric, quadratic programming are numerical methods wherein an approximate solution is sought by proceeding in an interactive manner by starting from an initial solution.
- ⑤ The Dynamic programming technique is useful for optimal control problems.
- ⑥ Stochastic programming deals with the variables described by probability distributions.
- ⑦ The modern methods of optimizations are - genetic algorithm, simulated annealing, particle swarm optimization, ant colony optimization etc.

Condition for minima or maxima of a function:-

Single variable function:-

Necessary Condition:- If a function $f(x)$ is defined in the interval $a \leq x \leq b$ and has a relative minimum / maximum at $x = x^*$ where

$a < x^* < b$

if the derivative $\frac{df(x)}{dx} = f'(x)$ exists as a finite no. at $x = x^*$ then $f'(x^*) = 0$

Sufficient Condition:-

Let $f'(x^*) = f''(x^*) = \dots = f^{(n-1)}(x^*) = 0$ but $f^n(x^*) \neq 0$

Then, $f(x^*)$ is - (i) a minimum value of $f(x)$, if $f^n(x^*) > 0$ and n is even.

- (ii) a maximum value of $f(x)$, if $f^n(x^*) < 0$ and n is even.

- (iii) neither maximum nor minimum if n is odd.

Determine the maximum and minimum values of the function.

$$f(x) = 12x^5 - 45x^4 + 40x^3 + 5$$

Since, $f'(x) = 60(x^4 - 3x^3 + 2x^2) = 60x^2(x-1)(x-2)$, $f'(x) = 0$ at $x=0$ and $x=2, x=1$

The second derivative is

$$f''(x) = 60(4x^3 - 9x^2 + 4x)$$

At $x=1$, $f''(x) = -60$ and hence $x=1$ is a relative maximum,

so, $f_{\max} = f(x=1) = 12$

At $x=2$, $f''(x) = 240$ and hence $x=2$ is a relative minimum,

so, $f_{\min} = f(x=2) = -11$

At $x=0$, $f''(x) = 0$ and hence, we must investigate the next derivative,

$$f'''(x) = 60(12x^2 - 18x + 4) = 240 \text{ at } x=0$$

Since $f'''(x) \neq 0$ at $x=0$, $x=0$ is neither a maximum nor a minimum and it is an inflection point.

Multivariable optimization with no constraints:-

Necessary Condition + If $f(x)$ has an extreme point at $x=x^*$ and if the partial derivatives of $f(x)$ exist at x^* , then

$$\frac{\partial f(x^*)}{\partial x_1} = \frac{\partial f(x^*)}{\partial x_2} = \dots = \frac{\partial f(x^*)}{\partial x_n} = 0$$

Sufficient Condition + A sufficient condition for a stationary point x^* to be an extreme point is that the matrix of second partial derivatives of $f(x)$ evaluated at x^* is —

- (i) positive definite when x^* is a relative minimum point.
- (ii) negative definite when x^* is a " maximum n.

$$J|_{x=x^*} = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]_{x=x^*}$$

is the matrix of second partial derivatives and is called the Hessian Matrix of $f(x)$.

In the case of a function of two variables, $f(x, y)$ the hessian matrix may be neither positive nor negative definite at a point (x, y) at which $\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0$. In such case, the point (x, y) is called a saddle point.

Multivariable optimization with equality constraints

Optimization of continuous functions subjected to equality constraints:

Minimize $f = f(x)$ subject to $g_j(x) = 0, j = 1, 2, \dots, m$

$$x = \begin{Bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{Bmatrix}$$

$n = \text{no. of variables}$

$m = \text{no. of equality constraints}$

By direct Substitution.

→ Solve simultaneously the m equality constraints and express any set of m variables in terms of the remaining $n-m$ variables.

→ A new objective function involving only $n-m$ variables, it is not subjected to any constraint and hence its optimum can be found by using the unconstrained optimization techniques.

→ Constraint equations will be linear for most of practical problems, and often it becomes impossible to solve them and express any m variables in terms of the remaining $n-m$ variables.

Find the solution using the lagrange multiplier method:

Minimize $f(x, y) = kx^{-1}y^{-2}$ subject to $g(x, y) = x^2 + y^2 - a^2 = 0$

The lagrange function is

$$L(x, y, \lambda) = f(x, y) + \lambda g(x, y)$$

$$= kx^{-1}y^{-2} + \lambda(x^2 + y^2 - a^2)$$

The necessary conditions for the minimum of $f(x, y)$ give

$$\frac{\partial L}{\partial x} = -kx^{-2}y^{-2} + 2x\lambda = 0 \rightarrow (1)$$

$$\frac{\partial L}{\partial y} = -2kx^{-1}y^{-3} + 2y\lambda = 0 \rightarrow (2)$$

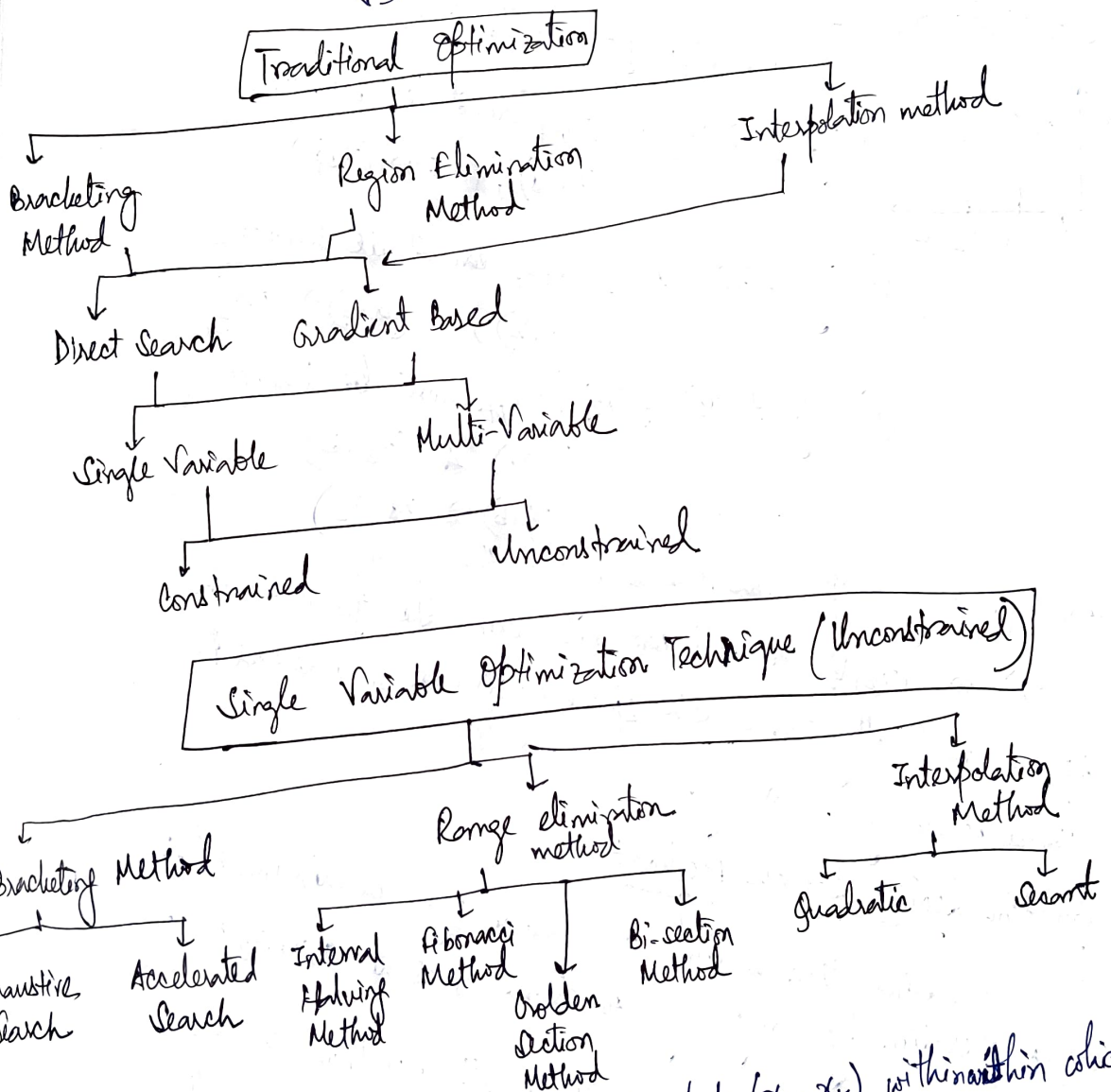
$$\frac{\partial L}{\partial \lambda} = x^2 + y^2 - a^2 = 0 \rightarrow (3)$$

from ① and ②

$$2\lambda = \frac{k}{x^3 y^2} = \frac{2k}{x y^4}$$

from which the relation $x^* = \left(\frac{1}{\sqrt{2}}\right) y^*$ can be obtained, This relation along with equation ③ gives the optimum solution as

$$x^* = \frac{a}{\sqrt{3}} \quad \text{and} \quad y^* = \sqrt{2} \frac{a}{\sqrt{3}}$$



Bracketing method $\rightarrow$ To identify a bracket $(x_1 - x_k)$ within which x_{opt} is present.

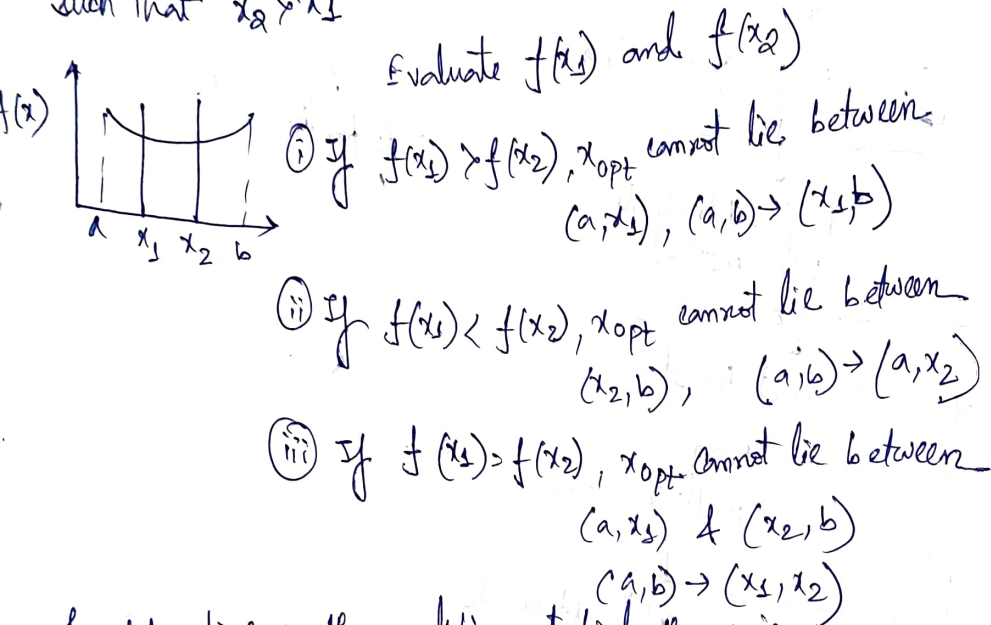
Region Elimination method $\rightarrow$ To identify a bracket $(x_1 - x_k)$ within which x_{opt} is not present. By eliminating the identified region the search space is gradually reduced to desired size.

Interpolation method $\rightarrow$ Curve fitting through iterative selection of points on the given curve.

Region Elimination method:-

To identify a bracket (x_1, x_2) within which x_{opt} is not present by eliminating the identified region the search space is gradually reduced to desired size.

Suppose that a $f(x)$, a unimodal function is defined on the interval (a, b) , select any two intermediate points x_1 and x_2 such that $x_2 > x_1$



So, depending on the condition satisfied the region (a, x_1) or (x_2, b) or both can be eliminated from search space. Gradually the search space (a, b) can be reduced to desired level of accuracy.

Golden Section Search Method:-

→ In region elimination method a selected region is eliminated in every iteration by comparing function values at a, x_1, x_2 and b .

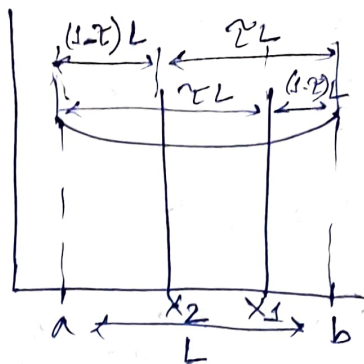
→ If one of the intermediate points in k th iteration remain valid intermediate point in $k+1$ th iteration, only one function evaluation required in each iteration.

→ The efficiency of these methods depend on the no. of function evaluation require before reaching to desired accuracy level.

Suppose $k=1$, $a_1=a$, $b_1=b$, $L_1=b_1-a_1$

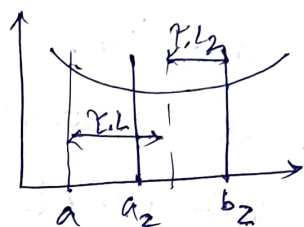
$$x_1 = a_1 + \tau \cdot L_1 \quad x_2 = b_1 - \tau \cdot L_1$$

If $f(x_1) < f(x_2)$ (a to x_2) to be eliminated.



$$k=2, a_2=x_2, b_2=b, L_2=b_2-a_2$$

$$= b - x_2 \\ = \tau \cdot L_1$$



$$k=2,$$

$$x_1 = a_2 + \tau \cdot L_2, \quad x_2 = b_2 - \tau \cdot L_2$$

$$= b - \tau \cdot (\tau \cdot L) = a + \tau \cdot L, \text{ then } x_1 \text{ at } k=1 \text{ will be } x_2 \text{ at } k=2$$

$$\text{If } b - \tau \cdot (\tau \cdot L) = a + \tau \cdot L$$

$$\Rightarrow (b-a) - \tau \cdot (\tau \cdot L) = \tau \cdot L \Rightarrow L - \tau^2 \cdot L = \tau \cdot L$$

$$\Rightarrow L(\tau^2 + \tau - 1) = 0 \Rightarrow (\tau^2 + \tau - 1) = 0 \Rightarrow \tau = 0.618$$

$$\Rightarrow L_n = (0.618)^{n-1} L_0$$

Bisection Method:-

Suppose that a $f(x)$ is a unimodal function defined on the interval (a,b)

$$1. x_m = (a+b)/2$$

2. Calculate derivatives $f_1(a)$, $f_1(b)$ and $f_1(x_m)$ at points a, b and x_m
check $f_1(a) \cdot f_1(b) < 0$

3. If, $f_1(a) \cdot f_1(x_m) < 0$, x_{opt} must lie between (a, x_m)

4. Hence $a = a$ and $b = x_m$ goto step 1

else $f_1(a) \cdot f_1(x_m) > 0$, x_{opt} must not lie between (a, x_m)

hence $a = x_m$ and $b = b$ goto step 1

In every iteration search space is halved and hence it is called bisection method.

Secant Method:-

In secant method,

suppose that a $f(x)$ is a unimodal function defined on the interval (a, b) .

1. Calculate derivatives $f_1(a), f_1(b)$ at points a, b

check $f_1(a) \cdot f_1(b) < 0$

2. Find x_m where $f_1(x_m) = 0$ (assume $f_1(x)$ varies linearly)

$$3. x_m = a + \left(\frac{-f_1(a)}{f_1(b) - f_1(a)} \right) \cdot (b - a)$$

4. If $f_1(a) \cdot f_1(x_m) < 0$, x_{opt} must lie between (a, x_m)
hence $a = a$, $b = x_m$ goto step 1

In Bisection method, sign of the derivatives are used to identify the region to be eliminated but in secant method value of the derivatives are used to guess the point where the value of the derivative is to be zero and then signs are used to identify the region to be eliminated.

Apply Interval halving method or Golden section method to find out minimum value of the objective function (2 iterations)

(10)

$$f(x) = x^2 + 25/x$$

Step 1:- We choose $a=0$ and $b=5$, The transformation equation becomes, $w = x/5$

Thus, $a_w = 0$, $b_w = 1$ and $L_w = 1$, Since the Golden section method works with a transformed variable w , it is convenient to work with the transformed ~~variable~~ function.

$$f(w) = 25x^2 + 25/(5x) = 25x^2 + 5/x$$

We set an iteration counter $k=1$.

Step 2:- We set $w_1 = 0 + (0.618) \cdot 1 = 0.618$

$$\text{and } w_2 = 1 - (0.618) \cdot 1 = 0.382$$

$$f(w_1) = 9.5481 + 8.09 = 17.6381$$

$$f(w_2) = 16.7371$$

Since $f(w_1) > f(w_2)$, the minimum cannot lie any point greater than $w = 0.618$.

Thus we eliminate the region (w_1, b) or $(0.618, 1)$

Thus $a_w = 0$, $b_w = 0.618$, At this stage, $L_w = 0.618 - 0 = 0.618$

Step 3:- Since $|L_w|$ is not smaller than ϵ , we set $k=2$ and move to Step 2. This completes one iteration of the golden section search method.

Step 2:- For the second iteration, we set,

$$w_1 = 0 + (0.618) \cdot 0.618 = 0.381924$$

$$w_2 = 0.618 - (0.618) \cdot 0.618 = 0.236076$$

$$f(w_1) = 16.7382, f(w_2) = 22.573$$

$$f(w_1) < f(w_2),$$

so, we eliminate the interval $(0, 0.236)$,

Thus, new bounds —

$$a_w = 0.236 \text{ and } b_w = 0.618,$$

the new interval is $L_w = 0.618 - 0.236 = 0.382$

which is incidentally equal to $(0.618)^2$

This completes the second iteration of the algorithm

Unconstrained Multivariable objective function:-

The objective funcⁿ for N variables represented by $x_1, x_2, \dots, x_N$ is $f(x) = f(x_1, x_2, \dots, x_N)$

The solⁿ of this funcⁿ is given by vector $x = [x_1, x_2, \dots, x_N]^T$

The optimum solⁿ of this funcⁿ is given by vector $x^* = \begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_N^* \end{bmatrix}$

Size of vector x , $|x| = \sqrt{x_1^2 + x_2^2 + \dots + x_N^2}$

Minimum value of $f(x) = f(x^*) = f(x_1^*, x_2^*, \dots, x_N^*)$

The gradient vector at any point x^t

$$\nabla f(x^t) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_N} \right)^T \bigg|_{x^t} \quad \left[\begin{array}{l} \text{The first order partial} \\ \text{derivatives can be} \\ \text{calculated numerically} \end{array} \right]$$

$$\frac{\partial f}{\partial x_1} = \frac{f(x_1 + \Delta x_1, x_2, \dots, x_N) - f(x_1, x_2, \dots, x_N)}{\Delta x_1}$$

In a multivariable funcⁿ, the gradient of a funcⁿ is not a scalar quantity, instead it is a vector quantity

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} \\ \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \vdots \\ \frac{\partial^2 f}{\partial x_N^2} \end{bmatrix}$$

The second order derivatives form a matrix, $\nabla^2 f(x^t)$.

Optimality Criteria:-

A point x is a stationary point if $\nabla f(x) = 0$. Furthermore, the point is a minimum, a maximum, an inflection point if $\nabla^2 f(x)$ is positive-definite, negative-definite or otherwise.

→ A matrix $\nabla^2 f(x)$ is defined to be positive-definite if for any point y in the search space the quantity $y^T \nabla^2 f(x) y \geq 0$

→ A matrix $\nabla^2 f(x)$ is defined to be negative-definite if for any point y in the search space the quantity $y^T \nabla^2 f(x) y \leq 0$.

Gradient-Based Methods:-

Gradient-based methods exploit the derivative information of the function and are usually faster search method. The derivatives can be obtained by numerical computation of objective functions only at two or three neighbouring points.

Steps → Most gradient-based methods work by searching along several directions iteratively, the algorithms vary according to the way the search directions are defined.

① While at point x^k , best search direction s^k is defined.

② A unidirectional search is performed along s^k to locate the minimum in this direction.

③ This is performed by first writing a representative point along the direction s^k as follows —

$$x^{k+1} = x^k + \alpha^k \cdot s^k$$

where α^k = step length,

④ Objective funcⁿ $f(x)$ will be a funcⁿ of one variable α .
⑤ Thus a unidirectional search for α_{opt}^k will identify best point along s^k and the new point x^{k+1} can be obtained. Thereafter, the search is continued from the new point x^{k+1} along another search direction s^{k+1} .

Convergence criteria $\rightarrow$

either $|x^{k+1} - x^k| \leq \epsilon_1$

or $|f(x^{k+1}) - f(x^k)| \leq \epsilon_2$

or both

let x^k be the current point at iteration level k

(Initial point for $k=1$)

best $s^k \rightarrow$ best possible search direction using funcⁿ or gradient values at x^k

(i) Cauchy's Steepest Descent Method :-

$$s^k = -\nabla f(x^k) ; x^{k+1} = x^k - \alpha^k \nabla f$$

This technique is effective when x^k is far away from x_{opt} .

(ii) Newton's Method :-

$$s^k = \frac{-\nabla f(x^k)}{\nabla^2 f(x^k)} ; x^{k+1} = x^k - \alpha^k \frac{\nabla f(x^k)}{\nabla^2 f(x^k)}$$

This technique is effective when x^k is very close to x_{opt}

(iii) Marquardt's Method

$$s^k = \frac{-\nabla f(x^k)}{(\nabla^2 f(x^k) + \lambda I)}$$

I = identity matrix
 λ = constant

λ starts with very large values, Cauchy's steepest descent
gradually reduced to low values $\therefore$ Newton's method

Cauchy's Steepest Descent Method:

Step 1 :- Choose a maximum no. of iterations M to be performed, an initial point $x^{(0)}$, two termination parameters ϵ_1, ϵ_2 and set $k=0$

Step 2 - Calculate $\nabla f(x^{(k)})$, the first derivative at point $x^{(k)}$

Step 3 - If $\|\nabla f(x^{(k)})\| \leq \epsilon_1$, Terminate;

Else if $k \geq M$, Terminate;

Else go to Step 4.

Step 4 - Perform a unidirectional search to find $\alpha^{(k)}$ using ϵ_2 such that $f(x^{(k+1)}) = f(x^{(k)} - \alpha^{(k)} \nabla f(x^{(k)}))$ is minimum

One criterion for termination is when $|\nabla f(x^{(k+1)}) \cdot \nabla f(x^{(k)})| \leq \epsilon_2$

Step 5 - Is $\frac{\|x^{(k+1)} - x^{(k)}\|}{\|x^{(k)}\|} \leq \epsilon_1$? If Yes $\rightarrow$ Terminate;

else get $k = k+1$ and go to Step 2.

Since the direction $s^{(k)} = -\nabla f(x^{(k)})$ is a descent direction, the function value $f(x^{(k+1)})$ is always smaller than $f(x^{(k)})$ for positive values of $\alpha^{(k)}$.

Marguardt's Method:-

Step 1: Choose the starting point, $x^{(0)}$, the max no. of iterations M and a termination parameter, ϵ .

Set $k=0$ and $\lambda^{(0)} = 10^4$

Step 2: Calculate $\nabla f(x^{(k)})$

Step 3: If $\|\nabla f(x^{(k)})\| \leq \epsilon$ or $k \geq M$? Terminate

Else goto Step 4.

Step 4: Calculate $s(x^{(k)}) = -[H^{(k)} + \lambda^{(k)}I]^{-1} \nabla f(x^{(k)})$

Set $x^{(k+1)} = x^{(k)} + s(x^{(k)})$

Step 5: Is $f(x^{(k+1)}) < f(x^{(k)})$? If yes goto Step 6 else go to Step 7.

Step 6: Set $\lambda^{(k+1)} = \frac{1}{2} \lambda^{(k)}$, $k = k+1$, goto Step 2.

Step 7: Set $\lambda^{(k)} = 2 \lambda^{(k)}$ and goto Step 4.

A large value of the parameter λ is used initially, thus, the Hessian matrix has little effect on the determination of the search direction. Initially the search is similar to that in Cauchy's method. After a no. of iterations (when the current solⁿ has converged close to the minimum), the value of λ becomes small and the effect is more like that in Newton's method. The algorithm can be made faster by performing an unidirectional search while finding the new point in Step 4.

Apply Cauchy's steepest descent method for multivariable optimization for the following function,
 $f(x) = (x_1 - 2x_2 - x_1x_2)^2$ (two iterations) (10)

Step 1 In order to ensure proper convergence, a large value of M (≈ 100) is usually chosen. The choice of M also depends on the available time and computing resource.
 let us choose $M=100$, an initial point $x^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}^T = (1, 1)$ and termination parameters $\epsilon_1 = \epsilon_2 = 10^{-3}$, we also set $k=0$

Step 2 $\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 2(x_1 - 2x_2 - x_1x_2) \cdot (1 - x_2) \\ 2(x_1 - 2x_2 - x_1x_2) \cdot (-2 - x_1) \end{bmatrix}$

$$\nabla f(x^{(0)}) = \begin{bmatrix} 0 \\ 12 \end{bmatrix}$$

Step 3 The magnitude of the derivative vector is not small and $k=0 < M=100$, Thus we do not terminate and proceed to

Step 4.

Step 4 Perform a line search from $x^{(0)}$ in the direction $-\nabla f(x^{(0)})$ such that the function value is minimum. Along that direction, any point can be expressed by fixing a value for the parameter $\alpha^{(0)}$ in the equation:

$$x = x^{(0)} - \alpha^{(0)} \nabla f(x^{(0)}) = (1, 1 - 12\alpha^{(0)})^T$$

$$f(1, 1 - 12\alpha) = (-2 + 36\alpha)^2$$

$$\text{Minimizing w.r.t. } \alpha \Rightarrow \frac{d}{d\alpha} (-2 + 36\alpha)^2 \Rightarrow \alpha = \frac{1}{18}$$

$$x^{(1)} = \left(1, \frac{1}{3}\right)^T$$

$$\nabla f(x^{(1)}) = \begin{bmatrix} 2(1-\frac{2}{3}-\frac{1}{3}) \cdot (1-\frac{1}{3}) \\ 2(1-\frac{2}{3}-\frac{1}{3}) \cdot (-2-1) \end{bmatrix} \\ = (0, 0)^T$$

$$\|\nabla f(x^{(1)})\| < 10^{-3}$$

Convergence achieved

Apply Bisection method (2 iterations) to find out optimal solⁿ for the following objective function. (5)

$$f(x) = x^2 + 5x + 3$$

$$\epsilon = 10^{-5}$$

Let's assume, $f(x)$ is defined on interval $(-5, 5)$

$$x_m = \frac{-5+5}{2} = 0$$

$$f'(x) = 2x + 5$$

$$f'(-5) = -10 + 5 = -5 ; f'(5) = 15 ; f'(0) = 5$$

$$f'(a) \cdot f'(b) < 0$$

$f'(a) \cdot f'(x_m) < 0$, x_{opt} must be between $(-5, 0)$

$a = -5$, $b = 0$, goto step 1

$$x_m = \frac{-5+0}{2} = -\frac{5}{2}$$

$$f'(-5) = -5 ; f'(0) = 5, f'(-\frac{5}{2}) = 0$$

$|f'(-\frac{5}{2})| < \epsilon$, so we terminate in iteration 2.

Using Golden section,

$$w = \frac{\lambda}{5}$$

$$k=1, \quad a_1 = -5, \quad b_1 = 5, \quad L_1 = 10$$

$$-1.$$

$$x_1 = a_1 + \gamma \cdot L_1 = -5 + 0.618 \times 10 = 1.18$$

$$x_2 = b_1 - \gamma \cdot L_1 = 5 - 0.618 \times 10 = -1.18$$

$$f(x_1) = 10.3$$

$$f(x_2) = -1.5$$

$f(x_1) > f(x_2)$, $(x_1 \text{ to } b)$ to be eliminated
 $[-5, 1.18]$

$$k=2, \quad a_2 = -5, \quad b_2 = 1.18, \quad L_2 = 1.18 + 5 = 6.18$$

$$x_1 = a_2 + \gamma \cdot L_2 = -5 + 0.618 \times 6.18 = -1.18076$$

$$x_2 = b_2 - \gamma \cdot L_2 = 1.18 - 0.618 \times 6.18 = -2.63924$$

$$f(x_1) = -1.5$$

$$f(x_2) = -3.23$$

$f(x_1) > f(x_2)$, $(x_1 \text{ to } b_2)$ to be eliminated

$$[-5, -1.18]$$

$$L_3 = -1.18 + 5 = 3.81924 = (0.618)^2 \times 10 \checkmark$$

Lagrange Multiplier for Multivariable Optimization and Equality Constraints

Consider the problem, minimize $f(x_1, x_2)$

subject to $g(x_1, x_2) = 0$

For this problem, the necessary condition for the existence of an extreme point $x = x^*$ was found to be

$$\left(\frac{\partial f}{\partial x_1} - \frac{\partial f / \partial x_2}{\partial g / \partial x_2} \frac{\partial g}{\partial x_1} \right) \bigg|_{(x_1^*, x_2^*)} = 0$$

by defining a quantity λ , called the Lagrange's multiplier, as

$$\lambda = - \left(\frac{\partial f / \partial x_2}{\partial g / \partial x_2} \right) \bigg|_{(x_1^*, x_2^*)}$$

$$\text{so, } \left(\frac{\partial f}{\partial x_1} + \lambda \frac{\partial g}{\partial x_1} \right) \bigg|_{(x_1^*, x_2^*)} = 0$$

$$\text{Similarly } \left(\frac{\partial f}{\partial x_2} + \lambda \frac{\partial g}{\partial x_2} \right) \bigg|_{(x_1^*, x_2^*)} = 0$$

In addition the constraint equation has to be satisfied at the extreme point, that $g(x_1, x_2) \big|_{(x_1^*, x_2^*)} = 0$

The necessary conditions given by these equations are more commonly generated by constructing a function L , known as the Lagrange's function as, $L(x_1, x_2, \lambda) = f(x_1, x_2) + \lambda g(x_1, x_2)$.

By treating L as a function of 3 variables x_1, x_2 and λ , the necessary conditions for its extremum are given by —

$$\frac{\partial L}{\partial x_1}(x_1, x_2, \lambda) = \frac{\partial f}{\partial x_1}(x_1, x_2) + \lambda \frac{\partial g}{\partial x_1}(x_1, x_2) = 0$$

$$\frac{\partial L}{\partial x_2}(x_1, x_2, \lambda) = \frac{\partial f}{\partial x_2}(x_1, x_2) + \lambda \frac{\partial g}{\partial x_2}(x_1, x_2) = 0$$

$$\frac{\partial L}{\partial \lambda}(x_1, x_2, \lambda) = g(x_1, x_2) = 0$$

In generalized case, it is defined by introducing one Lagrange's multiplier λ_j for each constraint $g_j(x)$

as

$$L(x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_m) = f(x) + \lambda_1 g_1(x) + \dots + \lambda_m g_m(x).$$

By treating L as a funcⁿ of the $n+m$ unknowns, the necessary conditions for the extremum of L , which also correspond to the solⁿ of the original problem —

$$\frac{\partial L}{\partial x_i} = \frac{\partial f}{\partial x_i} + \sum_{j=1}^m \lambda_j \frac{\partial g_j}{\partial x_i} = 0, \quad i=1, 2, \dots, n$$

$$\frac{\partial L}{\partial \lambda_j} = g_j(x) = 0, \quad j=1, 2, \dots, m$$

Minima:
$$Q = \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 L}{\partial x_i \partial x_j} dx_i dx_j > 0$$

Maxima:
$$Q = \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 L}{\partial x_i \partial x_j} dx_i dx_j < 0$$

Develop an NLP for multi-objective optimization problem:-

A rocket is designed to travel a distance of 12s in a vertically upward direction. The thrust of the rocket can be changed only at the discrete points located at distances of 0, s, 2s, 3s, ..., 12s.

If the maximum thrust that can be developed at point i either in the positive or negative direction is restricted to a value of F_i , formulate the problem of minimizing the total time of travel under the following assumptions:-

- (i) The rocket travels against the gravitational force.
- (ii) The mass of the rocket reduces in proportion to the distance traveled.
- (iii) The air resistance is proportional to the velocity of the rocket.

SOL:- Let control points on the path at which the thrust of the rocket are changed be numbered as 1, 2, 3, ..., 13, denoting x_i as the thrust, v_i the velocity, a_i the acceleration and m_i the mass of the rocket at point i .

By Newton's second law of motion -

net force on the rocket = mass $\times$ acceleration

$\Rightarrow$ Thrust - gravitational force - air resistance = mass $\times$ accel

$$\text{or, } x_i - m_i g - k_1 v_i = m_i a_i \quad \left[m_i = m_{i-1} - k_2 s \right]$$

where, k_1 and k_2 are constants,

$$a_i = \frac{x_i}{m_i} - g - \frac{k_1 v_i}{m_i}$$

If t_i denotes the time taken by the rocket to travel from point i to $i+1$, the distance traveled between the points

i and $i+1$ can be expressed as —

$$s = v_i t_i + \frac{1}{2} a_i t_i^2$$

$$m_i \frac{1}{2} t_i^2 \left(\frac{x_i}{m_i} - g - \frac{k_1 v_i}{m_i} \right) + t_i v_i - s = 0$$

$$\Rightarrow t_i = \frac{-v_i + \sqrt{v_i^2 + 2s \left(\frac{x_i}{m_i} - g - \frac{k_1 v_i}{m_i} \right)}}{\frac{x_i}{m_i} - g - \frac{k_1 v_i}{m_i}}$$

of the two values, the positive value has to be chosen for t_i .
The velocity of the rocket at point $i+1$, v_{i+1} can be expressed in terms of v_i as —

$$v_{i+1} = v_i + a_i t_i$$

$$v_{i+1} = \sqrt{v_i^2 + 2s \left(\frac{x_i}{m_i} - g - \frac{k_1 v_i}{m_i} \right)}$$

from an analysis of the problem, the control variables can be identified as the thrusts x_i , and the state variables as the velocities, v_i , since the rocket starts at point 1 and stops at point 13,

$$13, \quad v_1 = v_{13} = 0$$

Thus the problem can be stated as an OC problem as

Find $X = \left\{ \begin{matrix} x_1 \\ x_2 \\ \vdots \\ x_{12} \end{matrix} \right\}$ which minimizes.

$$f(X) = \sum_{i=1}^{12} t_i = \sum_{i=1}^{12} \left\{ \frac{-v_i + \sqrt{v_i^2 + 2s \left(\frac{x_i}{m_i} - g - \frac{k_1 v_i}{m_i} \right)}}{\frac{x_i}{m_i} - g - \frac{k_1 v_i}{m_i}} \right\}$$

subject to

$$m_{i+1} = m_i - k_2 s \quad i=1, 2, \dots, 12$$

$$v_{i+1} = \sqrt{v_i^2 + 2s \left(\frac{x_i}{m_i} - g - \frac{k_1 v_i}{m_i} \right)}$$

$$|x_i| \leq F_i \quad i=1, 2, \dots, 12$$

$$v_1 = v_{13} = 0$$

Box's Evolutionary Optimization Method

It is a simple optimization technique developed by G.E.P. Box in 1957.

The algorithm requires $(2^N + 1)$ points of which 2^N are corner points of an N -dimensional hypercube centered on the other point.

All $(2^N + 1)$ function values are compared and the best point is identified. In the next iteration another hypercube is formed around this best point. If at any iteration, an improved point is not found, the size of the hypercube is reduced. This process continues until the hypercube becomes very small.

ALGORITHM

Step 1 + Choose an initial point x^0 and size reduction parameters Δ_i for all design variables $i = 1, 2, \dots, N$. Choose a termination parameter ϵ .
Set $x = x^0$ ($x_1^0, x_2^0, \dots, x_N^0$)

Step 2 + If $\|\Delta\| < \epsilon$, Terminate;
else create 2^N points by adding & subtracting $\Delta_i/2$ from each variable at the point x^0 .

Step 3 + Compute function values at all $(2^N + 1)$ points, find the point having the minimum function value. Designate the minimum point to be x_{best} .

Step 4 + If $x_{\text{best}} = x^0$, reduce size parameters $\Delta_i = \Delta_i/2$ and goto Step 2.
else set $x^0 = x_{\text{best}}$ and goto Step 1.

(x^0 is always set as the current best point).
Thus at the end of simulation, x^0 becomes the obtained optimum point.